

NORTHEAST INDIA LOGISTICS RESILIENCE PLATFORM (NER-LRP)

AI & GIS-Driven Predictive Disruption Management & Resilient Logistics Control Tower

- **Problem Statement ID:** SIH26002
- **Problem Statement Title:** Predictive Logistics Decision Support & Disruption Management for NER
- **Theme:** Transportation & Logistics / Disaster Management / Smart Automation
- **PS Category:** Software (AI/ML + Geospatial GIS + Offline PWA + Multilingual)
- **Team ID:** [Your Team ID]
- **Team Name:** [Your Team Name - Registered on Portal]

Detailed Explanation of Proposed Solution

- **Sense-Predict-Decide-Act Control Tower:** Integrated regional command platform monitoring 4,820 km of critical mountain highways across all 8 Northeast states.
- **Real Live Data Ingestion:** Automatically ingests live precipitation & 48h rain from Open-Meteo/IMD APIs; integrates real NASA SRTM 30m slope gradients (e.g. 38.5° on NH-10).
- **AI Hazard Risk Engine:** Gradient Boosting ML model predicts 0-100 hazard scores with explainable factor contributions (Rain vs Slope vs Soil vs History).
- **Dynamic Route Re-planning:** Recalculates safe bypasses (e.g. NH-27 Haflong bypass around blocked NH-6 Sonapur) with exact ETA deltas.

How It Addresses Problem & Key Innovations

- **Cargo Criticality Weighting (Novel):** 85% safety priority for Emergency Medicines & Oxygen vs 25% for Commercial Bulk. Eliminates life-or-death stranding.
- **Offline-First Field Officer PWA:** Allows patrol officers in 0-connectivity mountain valleys to submit geo-photo reports cached in IndexedDB with auto-sync upon reconnect.
- **Multilingual Emergency Broadcaster:** Translates road warnings into 6 regional languages (Assamese, Mizo, Manipuri, Bengali, Hindi, English) with audible voice synthesis.
- **1-Click Disruption Simulator:** Enables live evaluators to trigger instant landslides (NH-6, NH-10, NH-29) and observe real-time rerouting in action.

Backend & GIS Pipeline

- Python FastAPI (Async REST API)
- Open-Meteo & IMD Live Weather Feeds
- SRTM DEM 30m Slope Gradient Data
- SQLite ACID DB with Spatial Schemas

AI / ML & Routing Engine

- Scikit-Learn Gradient Boosting ML
- Multi-variable Hazard Score (0-100)
- Dijkstra Multi-Objective Graph Router
- Cargo Priority Cost Functions

Frontend & Offline PWA

- React 18 + Vite + Tailwind Dark UI
- Leaflet GIS + Dynamic Map Layers
- IndexedDB Offline Local Storage Queue
- Web Speech Synthesis API (6 Languages)

End-to-End Implementation Methodology Workflow:

1. Live Ingestion: Open-Meteo queries rain/wind/soil moisture across 15+ NER stations + SRTM slope profiles → **2. ML Hazard Scoring:** GBDT model predicts 0-100 risk with explainable factors → **3. Cargo-Aware Routing:** Diverts medical shipments around risky segments with delta ETA → **4. Control Tower & Field PWA:** Visualizes live GIS map & syncs offline damage reports → **5. Multilingual Dispatch:** Broadcasts voice/text alerts across 6 regional languages.

Feasibility & Validation

- **Open Data Architecture:** Built entirely on accessible open data (Open-Meteo, OpenStreetMap, NASA SRTM). Zero licensing cost.
- **Working Prototype:** Fully functional full-stack app verified with 21 passing automated unit tests.
- **Corridor-Scale Focus:** Validated on critical lifelines: NH-10 (Sikkim), NH-6 (Meghalaya), NH-29 (Nagaland), NH-27 (Assam).

Potential Risks & Challenges

- **Govt Fleet Feeds Locked:** Real-time ULIP telematics require ministry authorizations.
- **Cellular Blackouts:** Mountain ghats frequently suffer zero 4G/5G mobile coverage.
- **Extreme Himalayan Terrain:** Sudden cloudbursts cause rapid slope failures.

Mitigation Strategies

- **Modular ULIP Adapter:** Standard GeoJSON interface ready to plug into ULIP when NDA is approved.
- **Offline-First PWA:** Caches reports in IndexedDB; auto-syncs idempotently upon signal recovery.
- **Hybrid Physics + ML:** Combines empirical IRC slope thresholds with GBDT machine learning.

Target Audience Impact • MDoNER & SDMAs: Unified regional command tower across all 8 Northeast states for coordinated emergency planning. • Hospitals: Prevents stock-out of emergency oxygen, anti-venom, and vaccines. • Drivers: Hands-free voice alerts prevent 48h mountain jam entrapment.	Economic Benefits • Inflation Suppression: Prevents local price spikes (>20%) in isolated districts. • Fuel & Delay Savings: Cuts vehicle turnaround times by 15-25% via planned detours. • Asset Protection: Protects multi-crore truck assets from sudden rockfall crushes.
Social & Environmental Benefits • Supply Reliability: Uninterrupted food grains & PDS rations to remote tribal hamlets. • Emission Reduction: Eliminates thousands of hours of diesel truck idling during blockades. • Grassroots Reporting: Empowers local scouts to report hazards directly.	System Resilience & PM GatiShakti • Emergency Corridors: Uncovers secondary state bypasses during national highway closures. • GatiShakti Alignment: Assists infrastructure planners in reinforcing high-risk slopes. • Himalayan Scalability: Replicable across Uttarakhand, HP, and J&K.;

Geospatial Data & Scientific Literature

- **Geological Survey of India (GSI):** National Landslide Susceptibility Mapping (NLSM) and Landslide Hazard Zonation (LHZ) protocols for Northeast India.
- **Academic ML Studies (Eastern Himalayas):** Gradient Boosting (XGBoost/LightGBM) models for landslide susceptibility in NE India (Dibang Valley & Nagaland, achieving AUC ~0.96 and ~0.89).
- **NASA SRTM 30m DEM:** High-resolution digital elevation data used for terrain slope and drainage gradient modeling across Himalayan highways.
- **Open-Meteo & IMD APIs:** Public meteorological endpoints providing live precipitation, 48h cumulative rainfall, and Doppler weather radar data.

Government Policies & Frameworks

- **PM GatiShakti National Master Plan:** Ministry of Commerce & Industry digital planning mandate for multimodal connectivity and logistics infrastructure resilience.
- **High-Level Task Force (HLTF) NER Report:** Official MDoNER reports detailing transport bottlenecks, logistics costs, and the 20% price inflation in remote hill states.
- **ULIP (Unified Logistics Interface Platform):** National Logistics Policy framework integrating multi-ministry vehicle telematics and digital transport documentation.
- **OpenStreetMap & OSRM Engine:** Open road graph datasets customized for heavy truck routing and slope penalty costs.